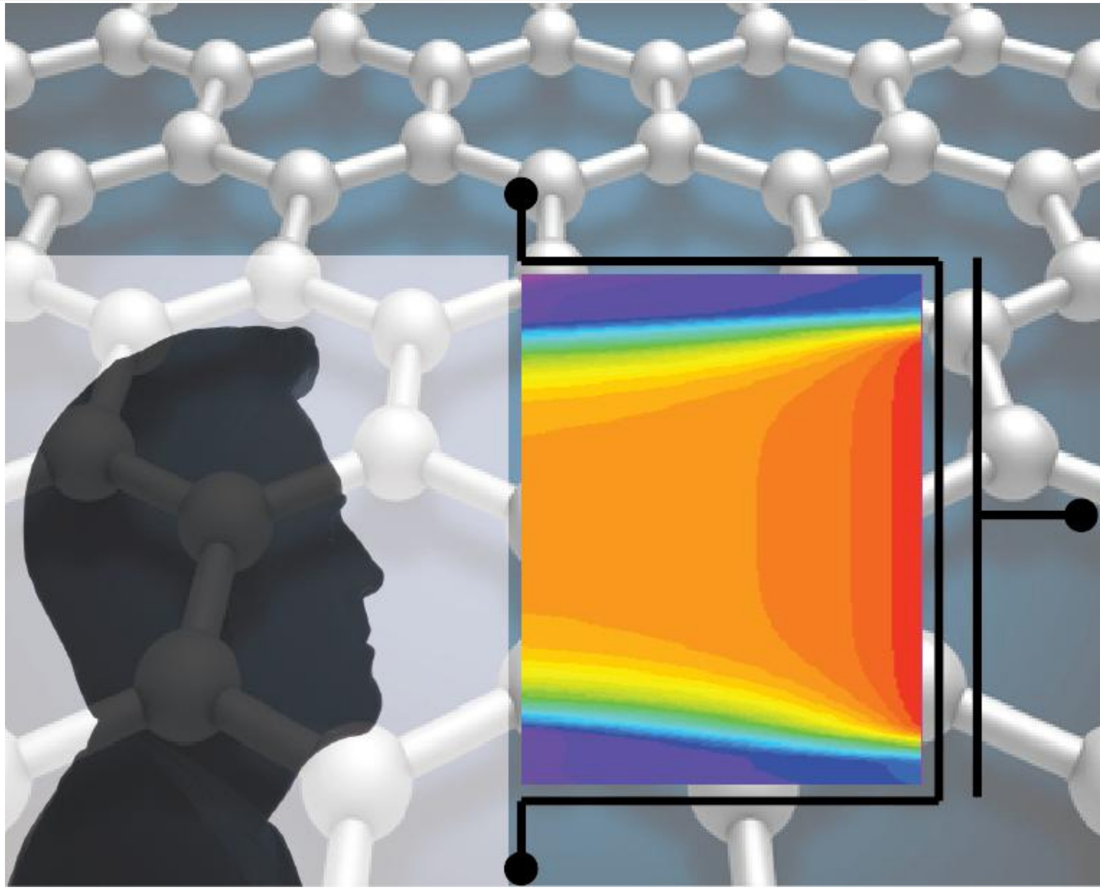
ADVANCED SOLID-STATE DEVICES

Week-12, Lecture-1



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7th April, 2025

MOSFET Scaling

MOSFET Scaling

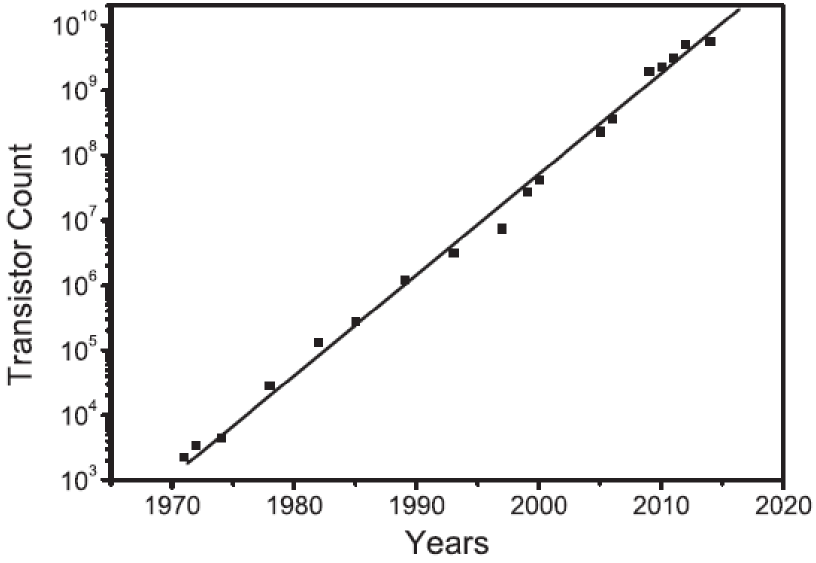
Basics

MOSFET Scaling: Concept

- *CMOS Scaling*: Reducing the *dimensions of the MOSFETs* and the wires connecting them in an IC such that the functionality of the IC remains unchanged

- Scaling theory proposed by ***Dennard*** et al. in 1974
- Basic operating characteristics of a MOSFET can be ***well-preserved*** if the parameters of the device are scaled in accordance with ***a given criterion***
- Scaling down of dimensions of transistors are done such that the ***geometric ratios*** that are important for the functioning of the IC remain unchanged

- Leads to exponential increase in the transistor count in integrated circuits



Increase in number of transistors for Intel's microprocessor

Source: S.Saurabh and M. J. Kumar, Fundamentals of Tunnel Field Effect Transistors, *CRC Press (Taylor & Francis)*, ISBN 9781498767132

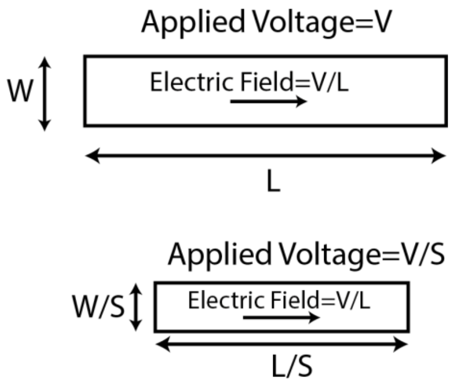
MOSFET Scaling: Types

Constant Field Scaling

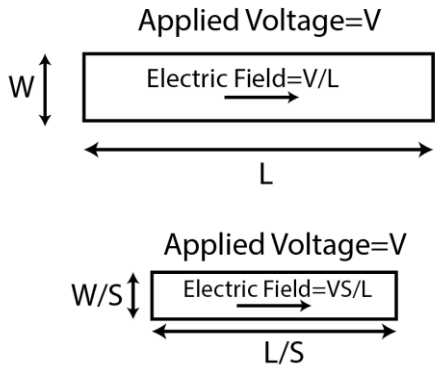
- Dimensions (*Length* and *Width*) of the MOSFET and the supply voltage are scaled by a factor S (where $S > 1$) such that the electric field in the MOSFET remains same
- This was proposed by Dennard and is also called *Dennard Scaling Rule*
- Now supply voltage cannot be scaled proportionately

Constant Voltage Scaling

- Dimensions (*Length* and *Width*) of the MOSFET are scaled by a S factor keeping supply voltage constant
- **Electric Field increase, potential reliability problems**



Constant Field Scaling



Constant Voltage Scaling

MOSFET Scaling: Motivation

Technology Nodes

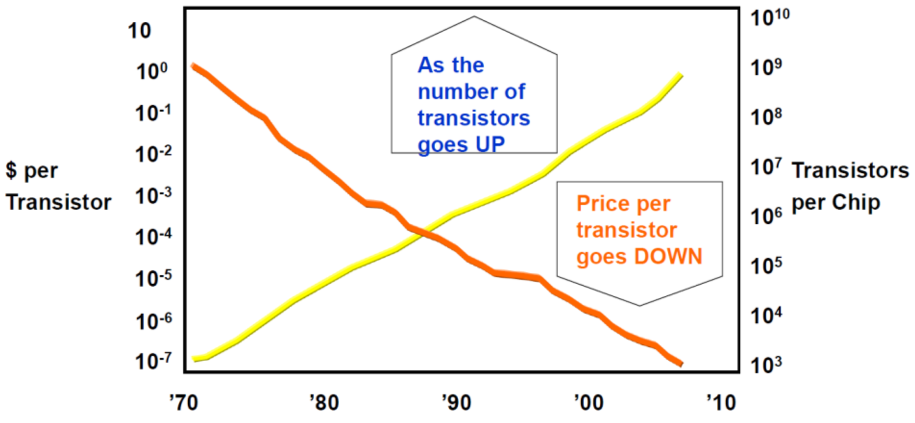
- 250 nm, 180 nm, 130 nm, 90 nm, 65 nm, 45 nm, 32 nm, ….

Moore's Law

- Results in decrease in Length and Width of the MOSFET by $S = \sqrt{2}$ in each successive generations of CMOS

Motivation for scaling (Improvement in device)

- The scaling of transistors as per Dennard's rule and Moore's law results in:
  - 100% increase in transistor density [more functionality]
  - 40% increase in the clock-frequency
  - 30% reduction in the power delay product [a measure of energy dissipated per switching operation]
  - No change in the area and the power dissipation of the circuit.



Source: *https://nanohub.org/resources/18348/download/Nikonov BeyondCMOS_1_scaling.pdf*

Motivation for scaling (Cost)

- Cost of fabrication remained fairly constant with scaling (depends primarily on the area of the circuit) ⇒ cost of fabrication per transistor decreases with scaling

Effect of Scaling

Problem:

Show that the speed of the circuit [frequency of the circuit] improves by around 40% in each subsequent generation, when Dennard's Scaling Law and Moore's law is assumed to be valid.

Assume that following equations hold true:

Capacitance $C = \frac{WL}{t_{ox}}$ Current $I = \frac{1}{t_{ox}}\frac{W}{L}V^2$

Circuit delay $\tau = \frac{CV}{I}$

Note these are approximate equations that can be used for first-order analysis to estimate the effect of scaling.

Solution:

Effect of Scaling

Comments:

- Assumptions used in the previous problem are not valid in latest technology nodes:
    - Impact of wire capacitance is dominant over gate capacitance
    - Velocity saturation must be taken into account
    - Dennard's Scaling Law is not valid [voltage cannot be scaled proportionately]
- Therefore, we no longer see large increase in clock frequency in subsequent generations of the circuit [for example: processors clock frequency is not increasing at 40%]

MOSFET Scaling

Challenges

Challenges to CMOS Scaling

Economic Challenges

- Keeping the ***manufacturing costs*** low, especially that of lithography at smaller device dimensions
  - Generally, cost advantage decreases in subsequent generations due to added complexity of fabrication
  - Unless the cost advantage is significant moving to next generation is not justifiable
- There should be a ***sustained demand*** for more silicon devices to provide the economic pull to Moore's Law
  - There should be pull from the market for more complex circuit with added features

Technical Challenges

- **Overall Power consumption**: Increasing trend has become a bottleneck to further increases in the complexity and the density of the circuits
  - ***Dark Silicon***: transistors are available, but cannot be used due to power constraints
- **Short channel effects**
- **Speed of circuit not increasing significantly**
- **Fabrication of smaller transistors**

MOSFET Scaling

Solutions

CMOS Scaling: Possible Solutions

Handling Short channel effects:

- FinFETs
- FDSOI FETs
- GAA: Nanosheet , Nanowires

Improving Speed of circuit:

- Strained Silicon: High mobility
- Novel materials such as III/V materials, CNT, graphene, TMD
- New interconnect materials: CNT, Graphene

Handling Power Constraints:

New devices with steep subthreshold swing:

- TFETs
- NCFETs
- NEMS

Advanced MOSFETs

Improvement in MOSFETs

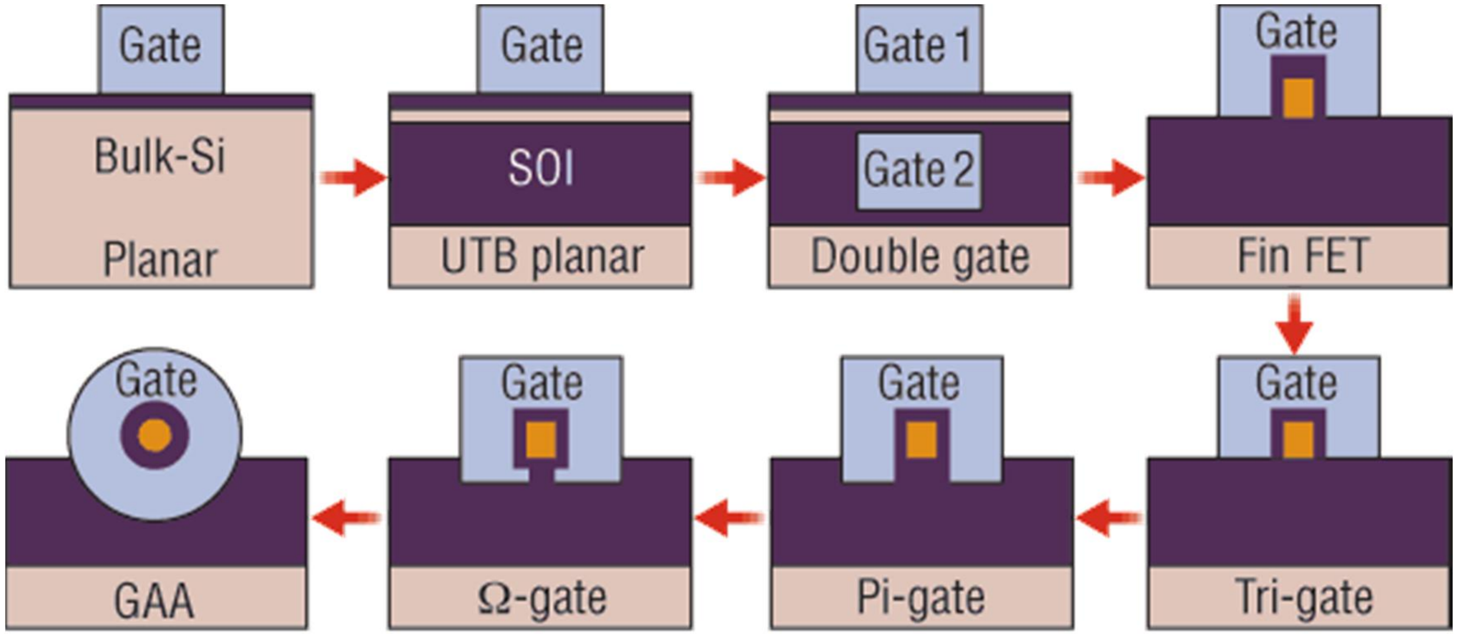
Channel potential $U_L = \alpha(-qV_D) + \beta(-qV_G)$ where V_D and V_G are drain and gate voltages and α, β quantify their relative contribution

Tackling Short Channel Effects:

- Reduce α and increase β
  - Tighter gate control compared to drain control over the channel potential
- Increase coupling of gate to the channel
- Use thinner channel
  - The part of channel away from the gate is responsible for leakage current and short channel effects
  - Remove that part of channel that is not well controlled by gate

How to achieve tighter gate control?

Gate Control: Evolution in Devices



Source: Liu, Tsu-Jae King. "FinFET history, fundamentals and future." *Proceedings of the Symposium on VLSI Technology Short Course*. University of California, 2012

- With advancement in MOSFET, the control of gate over channel goes on improving
- The Gate-All-Around (GAA) structure provides for the greatest capacitive coupling between the gate terminal and the channel

References

- S.Saurabh and M. J. Kumar, Fundamentals of Tunnel Field Effect Transistors, *CRC Press (Taylor & Francis)*, ISBN 9781498767132